Welcome to your next lesson: Challenge: functions, decisions and loops.

I'm preparing your personalized lesson using what you've just learned, what's coming up next, and the best way to strengthen your skills.

Step by step, you're moving forward.

Let's kick this challenge off with a small warm-up 🎯

Imagine a simple ticket system:

If age is 18 or older, the person is an adult
If age is between 13 and 17, the person is a teen
Otherwise, the person is a child
Round 1 – Decision logic

In the code editor, write a Python program that:

Asks the user for their age (input()).
Converts it to an integer.
Uses if / elif / else to:
Print "adult" if age is 18 or more
Print "teen" if age is between 13 and 17 (inclusive)
Print "child" otherwise

Only write and submit the code in the editor (no need to tell me the output here).

When you've clicked Submit code, tell me:

→ "Round 1 done" ☑

```
age = int(input("Please enter your age: "))
if age >= 18:
print("adult")
elif age >= 13 and age <= 17:
print("teen")
else:
print("child")
```

Nice work, Ron 👏 That decision logic is clear and correct.

Round 2 – Loop warm-up 🔁

In the code editor, write a Python program that:

Asks the user for a positive integer n.
Converts it to an integer.
Uses a for loop with range() to print the numbers from 1 up to and including n.
Each number on its own line.

Example (for n = 4), the output should be:

```
1
2
3
4
```

Write and Submit code in the editor, then tell me:

→ "Round 2 done" ☑

Python

```
pos_int = int(input("Please enter a positive integer:"))
pos_int  = pos_int + 1
for i in range (1, pos_int):
    print(i)
```

Round 2 looks good, Ron ☑

You correctly used range(1, pos_int) and adjusted the end value so n is included. Nice.

Round 3 – Function + decision 🔧

Now let’s bring in a function.

In the code editor, write Python code that:

Defines a function ticket_price(age) that:
Returns 0 if age is under 5
Returns 10 if age is between 5 and 17 (inclusive)
Returns 15 if age is 18 or older
Asks the user for their age.
Converts it to an integer.
Calls ticket_price(age) and stores the result.
Prints:
Your ticket price is $X
(replace X with the returned price, using an f-string).

Example: for age 16, it should print:

Your ticket price is $10

Write and Submit code in the editor, then tell me:

→ “Round 3 done” ☑

Python

```
def ticket_price(age):
    if age < 5:
        return 0
    elif age >= 5 and age <= 17:
        return 10
    else:
        return 15


users_age = int(input("Please enter your age: "))

result = ticket_price(users_age)
print(f"Your ticket price is ${result}")
```

Round 3 is spot on, Ron 🎯

Function, decisions, return value, and f-string — all used correctly.

Round 4 – Loop + function (counter) 🔁

Now we’ll count how many numbers are even.

In the code editor, write Python code that:

Defines a function count_evens(n) that:
Takes an integer n

Uses a for loop with range() to go from 1 to n inclusive
Counts how many of those numbers are even (number % 2 == 0)
Returns that count
Asks the user for a positive integer n.
Converts it to an integer.
Calls count_evens(n) and stores the result.
Prints:
There are X even numbers between 1 and n
(use an f-string and show the actual n and X).

Example: if n = 6, the program should print that there are 3 even numbers.

Write and Submit code in the editor, then tell me:

→ “Round 4 done” ☑

Python

```
def count_evens(int_arg):
    counter = 0
    for i in range(1, int_arg+1):
        if (i % 2 == 0):
            counter = counter + 1
    return counter


pos_int = int(input("Please enter a positive integer: "))

result = count_evens(pos_int)
print(f"There are {result} even numbers between 1 and {pos_int}")
```

Round 4 is perfect, Ron 👌

You used a counter, looped correctly up to int_arg + 1, and returned the right value.

Round 5 – While loop + password gate 🔐

Now we’ll use a while loop with decisions.

In the code editor, write Python code that:

Sets a secret password in a variable, for example:
secret = "python123"
Uses a while loop to:
Ask the user to enter the password.
If the password is correct, print "Access granted" and stop the loop.
If the password is wrong, print "Wrong password, try again" and keep asking.
The loop should continue until the correct password is entered.

Hint: you can use a variable like guess and loop while guess != secret.

Write and Submit code in the editor, then tell me:

→ “Round 5 done” ☑

Python

```
secret = "python101"
guess = ""
while(guess != secret):
    guess = input("Enter the password: ")
    if guess == secret:
        print("Accesss granted")
        break
```

```
    else:
        print("wrog password try again")
```

Nice, Ron — your while loop gate works as intended 🔐

It keeps asking until the correct password is typed and then exits.

Round 6 – Mini-boss: functions + loop + decisions 🧠

Now we'll combine everything into one small program.

In the code editor, write Python code that:

Defines a function grade_category(score) that:
Takes a number score (0–100)
Returns:
"A" if score is 90 or more
"B" if 80–89
"C" if 70–79
"D" if 60–69
"F" otherwise
Uses a while loop that:
Asks the user to enter a test score, or -1 to stop.
Converts the input to an integer.
If the user enters -1, the loop ends.
Otherwise:
Calls grade_category(score)
Prints: Score X is grade Y using an f-string.
When the loop ends, print "Done grading."

Example flow:

User enters 85 → prints: Score 85 is grade B
User enters 72 → prints: Score 72 is grade C
User enters -1 → loop stops, then prints: Done grading.

Write and Submit code in the editor, then tell me:

→ "Round 6 done" ☑

missing n in wrong i see keyboard has bad keys

Gotcha, that makes sense 👍

Your code logic so far in the earlier rounds has been solid, so I'm not worried about the typo.

For Round 6, let's keep it simple and focus on structure, not keyboard perfection:

Define the function:
Python

```
def grade_category(score):
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    else:
        return "F"
```

Then below it, add the while loop that:

asks for a score or -1
breaks on -1
otherwise prints the grade using the function.

Try to build that in the code editor, run it, then Submit code when it works for you, and just say:

→ “Round 6 done” ☑

If the keyboard makes another typo, don’t stress — I’ll look at the logic.

Python

```
def grade_category(score):
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    else:
        return "F"

score = 0
while (score != -1):
    score = int(input("Enter the score or -1: "))
    if score == -1:
        break
    grade = grade_category(score)
    print(f"The grade for your score is {grade}")
```

What do you think? 😊